

A HYBRID BLOCKCHAIN-AI FRAMEWORK FOR SECURE AND INTELLIGENT CONFIDENTIAL DOCUMENT MANAGEMENT

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Submitted in partial fulfillment of the requirements of the degree of

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Department of Statistics and Computer Science

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DECLARATION

I hereby declare that except where specific reference is made to the work of others, the contents of this dissertation are original and have not been submitted in whole or in part for consideration for any other degree or qualification in this, or any other university. This dissertation is my own work and contains nothing which is the outcome of work done in collaboration with others, except as specified in the text and Acknowledgements. This dissertation contains fewer than 65,000 words including appendices, bibliography, footnotes, tables and equations and has fewer than 150 figures.

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D. M. Y. W. S. Dissanayake
November 26, 2025

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November 26, 2025

ABSTRACT

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This project addresses the challenge of securely managing confidential documents by proposing a Hybrid Blockchain–Artificial Intelligence(AI) Framework designed for secure storage, intelligent retrieval, and verifiable validation of sensitive records. Traditional document systems rely on centralized storage, making them vulnerable to tampering, unauthorized access, and inefficient manual verification. The proposed solution integrates blockchain immutability, encrypted InterPlanetary File System(IPFS) storage, role-based access control, and agentic AI reasoning to deliver a transparent and auditable document management ecosystem. A hybrid storage model is implemented in which confidential documents are encrypted and stored off-chain on IPFS, while document metadata and integrity hashes are recorded on an EVM-compatible blockchain. A role-based validation mechanism models structured approval workflows, ensuring ordered document progression and preventing unauthorized or out-of-sequence actions across institutional roles. To enable intelligent interaction with stored records, the system introduces a Retrieval-Augmented, agentic AI layer built using LangGraph. The agent retrieves on-chain metadata, decrypts off-chain documents, and produces explainable responses with explicit reasoning traces. Experimental evaluations demonstrate reliable consistency between blockchain and IPFS content, 100% correctness in validator workflows, improved gas efficiency, and an AI query accuracy of 97.3%. Although validated using university document workflows, the framework is usable across domains and applicable to healthcare, legal, and administrative environments that require secure storage, transparent approval chains, and intelligent document extraction. The results show that combining blockchain, decentralized storage, and agentic AI provides a strong and scalable foundation for next-generation confidential document management.

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CHAPTER 1

INTRODUCTION

1.1 Introduction

The management of sensitive university documents like academic transcripts, registration records, recommendation letters, reference letters, research publications and meeting minutes face so many challenges within traditional centralized systems. In a traditional system, these are vulnerable to security risks such as unauthorized access, data tampering, and single point of failure which can be a threat to the integrity and reliability of critical academic and administrative records. Additionally centralized systems are lack of transparent and immutable tracking of history of documents, making verification inefficient and exposing the system to document forgery and disputes.

Consider a typical scenario: a student applies for a scholarship requiring verified academic transcripts. In the current system, the student submits a request to the examination division, which must manually verify records, print documents, obtain signatures from the Head of Department, Dean, Assistant Registrar, and Deputy Vice-Chancellor in sequence, that's a process taking 2-3 weeks. During this time, documents physically move between offices, creating opportunities for loss, unauthorized alterations, or forgery. The requester has no visibility into approval status, and if a document is questioned later, there is no tamper-proof audit trail showing who approved what and when. Similar inefficiencies plague land registries (where property title transfers require manual verification of ownership chains) and vehicle departments (where registration certificates lack verifiable provenance).

The core problems are:

1. Lack of Immutability: Documents can be altered after issuance without detection
2. No Transparent Audit Trail: Approval histories are not publicly verifiable
3. Poor Accessibility: Retrieving specific information requires manual searching through records

These challenges are not limited to universities. Government land registries and vehicle departments handle various document types like title deeds, ownership transfers and registration certificates that face the same risks in centralized setups. People can change files without permission, they can be altered as they pass between offices, no one sees a clear audit trail across departments, and one outage can stop everything.

Therefore my work targets a reusable document-assurance platform that works on different domains. The core design stays domain-neutral: smart contracts keep public facts (IDs, hashes, links/CIDs) and versions with role rules; IPFS stores the encrypted files; and an AI agent reads from both to answer questions and flag order problems. In the university context this enforces $\text{HOD} \rightarrow \text{Dean} \rightarrow \text{AR} \rightarrow \text{DVC}$ for letters; in land/vehicle contexts the same rails support chains like $\text{Notary} \rightarrow \text{Registrar} \rightarrow \text{Commissioner}$ (or $\text{seller} \rightarrow \text{buyer} \rightarrow \text{DMV}$), giving both settings transparent history, versioned updates, and easy verification.

While blockchain technology has appeared as a solution for secure document storage through its decentralized, immutable and transparent nature, it has its own set of challenges too. Current blockchain-based document management systems inefficient at data retrieval, making it difficult to quickly access stored information. To retrieve some data, users must manually search through blockchain records, which is time-consuming and impractical for large-scale document repositories.

The integration of Artificial Intelligence (AI) with blockchain technology can enhance the automation, intelligence, and trust in decentralized systems. In this approach, AI can seamlessly interact with blockchain networks by accessing, extracting, summarizing and interpreting on-chain data to support advanced functionalities. Sensitive information like grades of a student is securely stored off-chain (to protect user privacy) while blockchain maintains verifiable links to this off-chain data. The AI system is capable of answering domain-specific questions related to the university document registration and administrative workflows by using both on-chain and off-chain data. Additionally, for information not available within the system, the AI performs real-time internet searches to provide relevant answers.

1.2 Research Aim and Objectives

Research Aim:

The aim of this research is to develop a Hybrid Blockchain-AI framework for secure and intelligent management of confidential documents, combining blockchain for tamper-evident storage with AI for reasoning, retrieval, and workflow enforcement.

Research Objectives:

1. To develop a secure storage mechanism for confidential documents on blockchain.
2. To develop an Agentic document extraction system with clear reasoning, tool use, and an integrated Retrieval-Augmented Generation (RAG) system.
3. To develop validator-driven, role-based workflows for processes such as registration, issuance, and transfer.
4. To design a RAG system for monitoring and analyzing system workflows.

1.3 Research Questions

The study is guided by the following research questions:

1. How can blockchain ensure secure and immutable document storage?
2. What AI techniques, including Agentic Document Extraction (ADE), can enable intelligent document retrieval?
3. How can AI-driven fraud detection enhance the security of document management systems?
4. What are the challenges and limitations of a Blockchain-AI hybrid framework?

CHAPTER 2

LITERATURE REVIEW

Blockchain technology has many uses in the context of document management, especially because it allows for decentralized, transparent and immutable storage. One of blockchain's biggest advantages is that once a record is added, it cannot be changed or tampered with. Data on the blockchain is protected through cryptographic hash functions, digital signatures, and consensus mechanisms. Each block is cryptographically linked to its predecessor, forming an auditable and secure chain [1]. This ensures the integrity and authenticity of important documents. This is very useful for sensitive records like property deeds or healthcare files, where keeping reliable and verifiable history is essential [2] [3]. Blockchain uses smart contracts. Also known as self-executing programs that automate tasks like access control. However, finding specific data within a large blockchain can be difficult. While the ledger safely stores records, searching for certain content can be complex and slow. One common solution is connecting blockchain with an off-chain storage system like InterPlanetary File System (IPFS), which helps to manage large amounts of data [3]. IPFS ensures content integrity through content-addressing and stores encrypted files in a decentralized manner, with blockchain recording the hash for immutability and auditability [4].

Several recent studies have explored blockchain-based solutions for document storage and authentication. Dighe et al. (2024) proposed "DocBlock" [5], a permissioned blockchain framework designed to securely store and authenticate digital documents. Their approach records cryptographic hashes of documents on-chain while keeping the original files off-chain, ensuring both privacy and immutability. The system implements

access control mechanisms and digital signatures to prevent unauthorized access and document forgery. Although the authors suggest integrating artificial intelligence and machine learning as future enhancements, the framework remains primarily focused on blockchain-driven security and verification. The present research extends beyond this by combining blockchain's immutability with AI-based reasoning and Retrieval-Augmented Generation (RAG) pipelines to enable intelligent document retrieval and workflow analysis [5].

AI agents are autonomous software entities capable of adapting, learning, and executing multi-step operations. Unlike traditional bots, they can dynamically interact with protocols, external data streams, and human communities without direct human intervention [6]. The combination of blockchain and artificial intelligence creates a powerful system for managing documents by merging blockchain's security with AI's advanced data processing abilities [7]. AI can automate document extraction, classify information, interpret legal texts, and even improve how users interact with smart contracts using natural language [2]. NLP functionalities in AI systems enable users to interact with smart contracts using everyday language, making them more accessible and reducing the need for legal expertise [2].

Various case studies showcase the use of integrated Blockchain-AI systems across different industries, even if not always directly focused on document management. In finance, AI combined with blockchain enhances security through fraud detection, identity verification, and smart contract automation, ensuring tamper-proof transactions and regulatory compliance [8]. Supply chains benefit from AI's predictive analytics and real-time monitoring alongside blockchain's secure tracking of goods and contract enforcement [7]. In healthcare, AI analyzes patient data stored securely on blockchain, improving diagnosis and treatment while maintaining privacy and consent management [2].

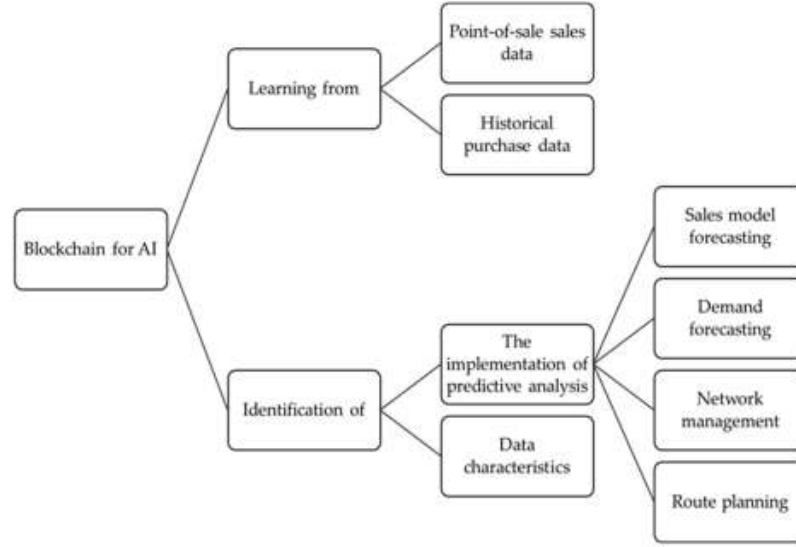


Figure 2.1: The applications of integrating AI and Blockchain in supply chain.

Meanwhile, blockchain's decentralized and tamper-proof ledger ensures that documents cannot be altered or forged, preventing fraud such as duplication or unauthorized modifications [9]. Validating integrated Blockchain-AI systems requires ensuring data integrity and AI reliability. Blockchain inherently maintains integrity through its decentralized ledger, cryptographic hashing, and consensus mechanisms that prevent tampering and ensure agreement on new data. Off-chain document verification using hash values further secures data accuracy [10].

Recent advancements have further extended blockchain-based document management beyond academic and land registry applications. Dinde and Shirgave (2024) introduced the "Legal eVault" framework [11], which utilizes a permissioned blockchain (Hyperledger Fabric) and the InterPlanetary File System (IPFS) for secure and decentralized management of legal documents. Their system emphasizes document immutability, encryption, and a Role-Based Access and Security (RBAS) model to ensure that only authorized stakeholders can upload or retrieve records. While their framework effectively enhances storage security and auditability in legal contexts, it primarily focuses on the blockchain layer without incorporating intelligent reasoning or retrieval mech-

anisms. The current study builds upon such foundational work by integrating an AI-driven reasoning layer and Retrieval-Augmented Generation (RAG) pipelines, enabling autonomous document understanding and validation across multiple institutional domains [11].

Integrating AI agents with blockchain provides enhanced transparency, security, and decentralized workflow automation. AI agents facilitate tasks such as role-based access control, validation, and policy enforcement, while on-chain storage of document metadata and off-chain encrypted storage (IPFS) support scalable and tamper-proof record management which is crucial for confidential applications like university records and certificates [12].

Implementing hybrid Blockchain-AI frameworks for secure document management requires addressing scalability, processing speed, and regulatory compliance. Blockchain’s limited transaction capacity and AI’s high data processing needs pose scalability challenges, which can be mitigated with improved consensus mechanisms, sharding, and layer-two scaling solutions[7]. Legal and privacy considerations are crucial when handling sensitive documents, as blockchain’s transparency risks exposing private data. Encryption techniques help secure information. AI decisions must also be transparent and explainable, with mechanisms to audit decision-making processes [2].

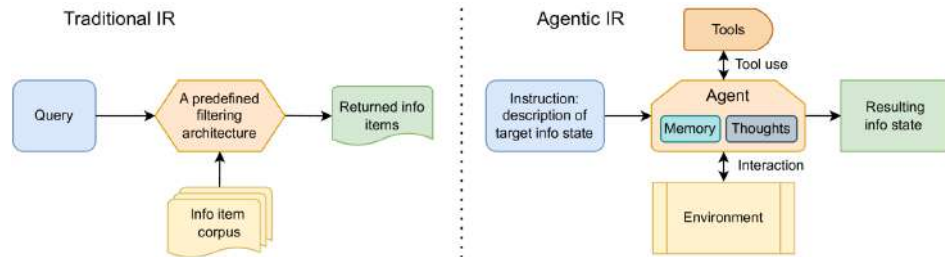


Figure 2.2: The paradigms of traditional IR vs. agentic IR.

Agentic Document Extraction (ADE) enhances the integration of AI and blockchain in secure document management by introducing autonomous, intelligent agents capable of extracting, structuring, and interpreting data. The autonomous agents in ADE systems can adapt to different document formats, learning from contextual cues and prior extractions to improve accuracy and efficiency over time [13].

This approach is particularly beneficial in legal and administrative applications, where structured document formats such as land deeds and vehicle registrations require precise extraction and validation of ownership records. When combined with blockchain, ADE ensures that extracted metadata is stored in an immutable, decentralized ledger, preserving document authenticity while enabling rapid searchability. Additionally, ADE-driven AI workflows can interact with smart contracts to verify extracted data against predefined conditions. The combination of blockchain's immutable storage and ADE's intelligent extraction capabilities significantly enhances document security, retrieval efficiency, and compliance with regulatory standards.

Table 2.1 provides a comparative analysis of existing blockchain-based document management systems against the proposed framework. While DocBlock and Legal eVault establish foundational approaches for secure document storage using blockchain and IPFS, they lack intelligent retrieval mechanisms and advanced workflow automation. The proposed system extends these concepts by integrating agentic AI reasoning, dynamic role-based workflows, and explainable document intelligence, positioning it as a next-generation solution for confidential document management across multiple institutional domains.

Table 2.1: Comparison of Blockchain-Based Document Management Systems

Feature	DocBlock [5]	Legal eVault [11]	Proposed System
Blockchain Type	Permissioned	Hyperledger Fabric	EVM-compatible (Tenderly Sepolia)
Off-chain Storage	Not specified	IPFS	IPFS with envelope encryption
Access Control	Digital signatures	Role-Based Access & Security (RBAS)	RBAC + ACL-based key wrapping
Encryption	Hash-based verification	Standard encryption	Envelope encryption with per-principal key wrapping
AI Integration	Suggested as future work	None	Agentic AI with LangGraph & RAG
Workflow Management	Basic verification	Document upload/retrieval	Dynamic hierarchical validation with state persistence
Intelligent Retrieval	None	None	Natural language queries with user-aware filtering
Workflow Monitoring	None	None	RAG-based analytics & bottleneck detection
Document Versioning	Not specified	Not specified	Blockchain-based immutable versioning
Query Transparency	N/A	N/A	Explainable reasoning traces
Primary Domain	General documents	Legal documents	Academic records (adaptable to other domains)

CHAPTER 3

METHODOLOGY

3.1 Overview

Although the implementation described in this thesis is demonstrated using a university records management scenario (transcripts, letters,..), the methodology is not limited to this domain. The choice of university documents was primarily for ease of testing, availability of safe sample data, and accessibility of stakeholders for feedback. However, the underlying architecture, combining blockchain, off-chain storage, role-based access, and agentic AI, is generic and can be applied to other sensitive domains such as land registries, vehicle, and healthcare records.

This research uses a hybrid approach that combines blockchain and artificial intelligence (AI) to manage confidential documents. The system has two parts:

1. On-chain (Blockchain) - Used for public metadata, immutability, and audit trails.
2. Off-chain (AI + IPFS) - Used for storing private documents securely and supporting intelligent queries.

The blockchain is deployed on Ethereum's EVM (Tenderly Sepolia) network for testing smart contracts. Confidential documents are encrypted and stored in IPFS, while blockchain stores only metadata and cryptographic proofs. This setup provides both transparency and security.

3.2 System Architecture

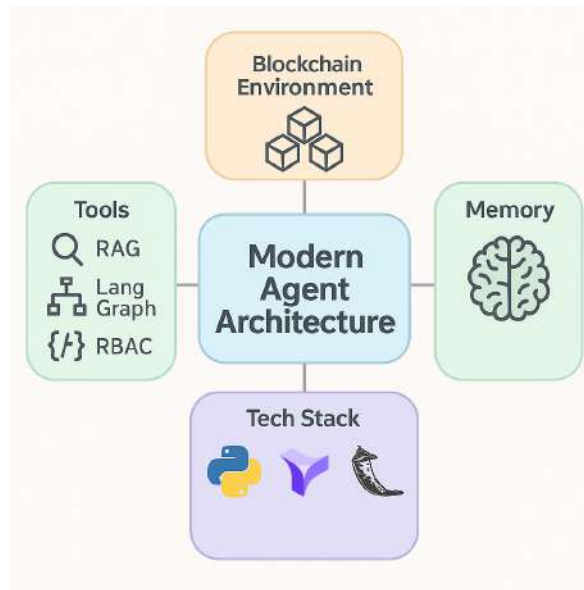


Figure 3.1: High-level Hybrid Blockchain-AI Workflow

The system is divided into three main components:

1. Blockchain Client

- Registers student records and stores references to their documents.
- Adds new semester records when required.
- Retrieves student details and semester document hashes.
- Key functions include:
 - Registering a student.
 - Adding a semester record.
 - Getting student details.
 - Getting all semester hashes.

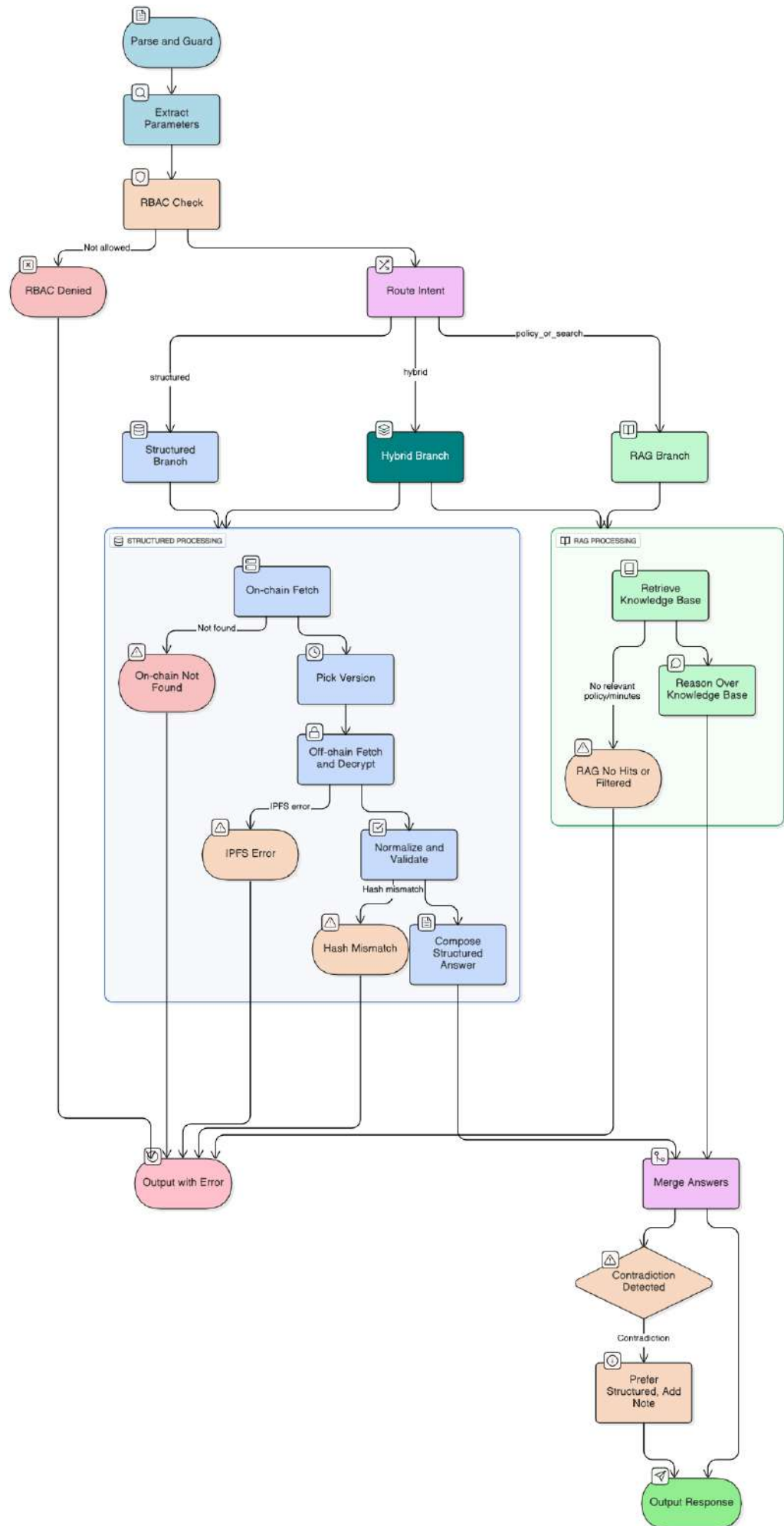
2. IPFS Client

- Handles document encryption and storage.

- Ensures confidentiality by encrypting documents before saving them.
- Provides functions to store, retrieve, and verify academic documents.
- Uses SHA-256 hashing to confirm integrity of the document.

3. AI Agent

- Uses LangGraph to confirm integrity of the document.
- Steps include:
 - Getting student details from blockchain.
 - Retrieving semester documents from IPFS.
 - Combining them and generating an answer through Llama3 model.
- This allows the system to support natural language questions like “Show semester 2 grades for STD001”.



The overall system architecture is based on a hybrid blockchain model where public metadata is stored on the blockchain and confidential files remain in IPFS. This structure provides a strong balance between security and performance. The blockchain part is deployed on Ethereum's EVM (Tenderly Sepolia) network, which was chosen for testing and simulation because it supports smart contracts and easy integration through Web3 libraries.

When a student's document is uploaded, the system first encrypts the file using a symmetric encryption key(Fernet symmetric encryption). Before uploading any document to IPFS, the system encrypts its content using the Fernet symmetric encryption algorithm. This ensures confidentiality even though IPFS is a public distributed storage network.

```
1 from cryptography.fernet import Fernet
2
3 key = Fernet.generate_key()
4 cipher = Fernet(key)
5
6 original = '{"student_id": "STD001", "grades": {"CSC1023": "A"}}'
7
8 encrypted = cipher.encrypt(original.encode())
9 decrypted = cipher.decrypt(encrypted).decode()
10
11 print(decrypted)
```

The encrypted file is then uploaded to IPFS, which returns an IPFS hash. This hash, along with other metadata such as the student's ID, program, and year, is then stored on the blockchain through a smart contract transaction. This design ensures that the document's authenticity can always be verified by checking if the document's hash matches the one stored on-chain. The architecture therefore includes several key layers:

1. Blockchain Layer – Handles all on-chain storage, transactions, and smart contract

executions.

2. IPFS Layer – Provides decentralized off-chain storage for encrypted confidential documents.
3. AI Layer – Integrates the LangGraph agent for intelligent query handling and fraud detection.
4. Application Layer – Provides the web-based user interface and backend logic for system operations.

This layered architecture provides a modular and scalable setup where each component can evolve independently while still maintaining a secure and consistent overall framework.

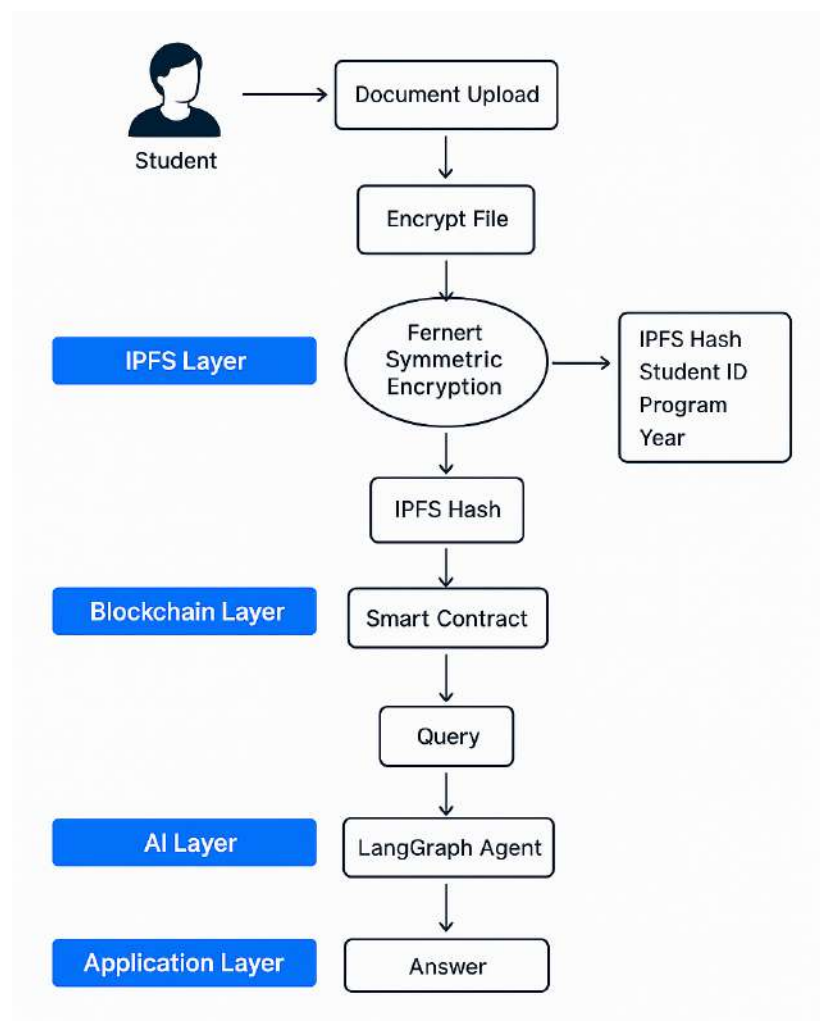


Figure 3.3: Hybrid Blockchain-AI Document Management System Architecture

3.3 Smart Contract Layer

The smart contract layer is the core of the blockchain component. It defines and manages all critical workflows related to academic document storage and verification. These main contracts were developed: `registerStudent`, `addSemesterRecord`, `getStudentDetails`, and `getStudentSemesterHashes`.

The `registerStudent()` function is used to create a new student record on the blockchain. It stores essential metadata such as the student's ID, name, academic program, current year, and a unique wallet address. Each student thus has a unique digital identity within the blockchain network. The `addSemesterRecord()` function is used whenever a new academic document (for example, a semester transcript) is uploaded. This function stores the IPFS hash and a cryptographic proof of the document's content. This adds new semester or letter documents with immutable versioning. `getStudentDetails()` and `getStudentSemesterHashes()` functions retrieve records and version histories. Together, these contracts ensure that all important student-related data is securely recorded and cannot be modified after submission. Each modification creates a new immutable record, preserving the document's historical chain. This ensures full version traceability.

```
1 student_data = {  
2     "student_id": "STD001",  
3     "name": "Yasela Dissanayake",  
4     "program": "BSc (Hons) Computer Science",  
5     "year": 4  
6 }  
7  
8 ipfs_hash = "QmZ9..."  
9 content_hash = "a4f2..."  
10 tx_hash = blockchain_client.register_student_record(student_data, ipfs_hash,  
    ↪ content_hash)
```

```
11 print(f"Registered student on blockchain. Transaction: {tx_hash}")
```

The blockchain client interacts with Ethereum-compatible nodes through Web3.py, allowing Python-based systems to invoke smart contract functions seamlessly.

In addition, each blockchain transaction includes a timestamp and a hash of the uploaded content. This acts as a verifiable audit trail that records who uploaded what, and when. Any future verification simply involves comparing the hash of the document with the one stored on the blockchain. If the two hashes match, it means the document is authentic and has not been altered since it was first recorded.

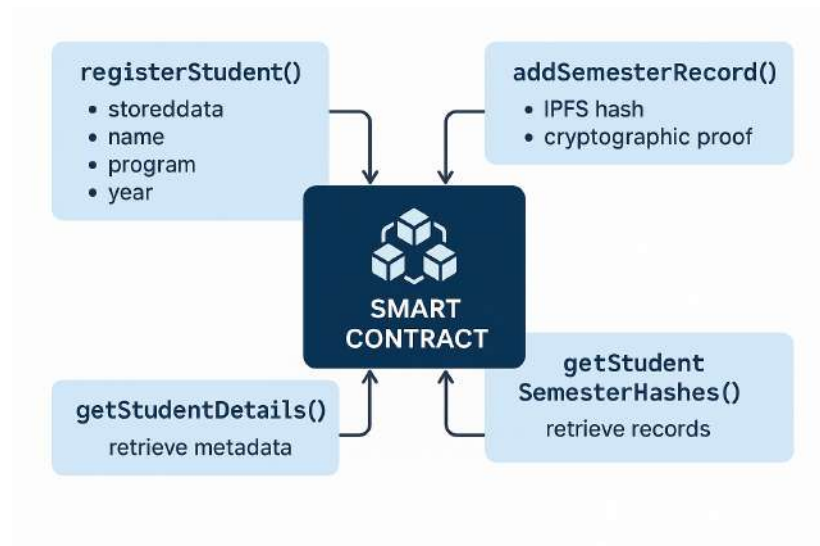


Figure 3.4: Smart contracts in document flow

3.4 Application Layer (Flask Frontend + Python Backend)

The application layer is built using the Flask framework for the web interface and a Python backend for the main system logic. This layer connects users with the blockchain and IPFS systems through simple, user-friendly web pages. The main web pages include features for registering new students, uploading semester documents, AI

query, admin panel for role management, login page, and validator dashboard for validators.

Authenticated sessions use Flask's session variables:

```
session["user"] = user["username"]
session["role"] = user["role"]
```

This context secures every route and enforces role-based access across the UI.

The Python backend is responsible for handling all the logic related to blockchain and IPFS. It uses the Web3.py library to interact with Ethereum's smart contracts and IPFS HTTP client to upload and retrieve files. When a document is uploaded, the backend encrypts it, adds it to IPFS, and then calls the blockchain contract to store the hash. Similarly, when a query is made, the backend fetches the data from both blockchain and IPFS, decrypts the content, and sends it back to the frontend for display.

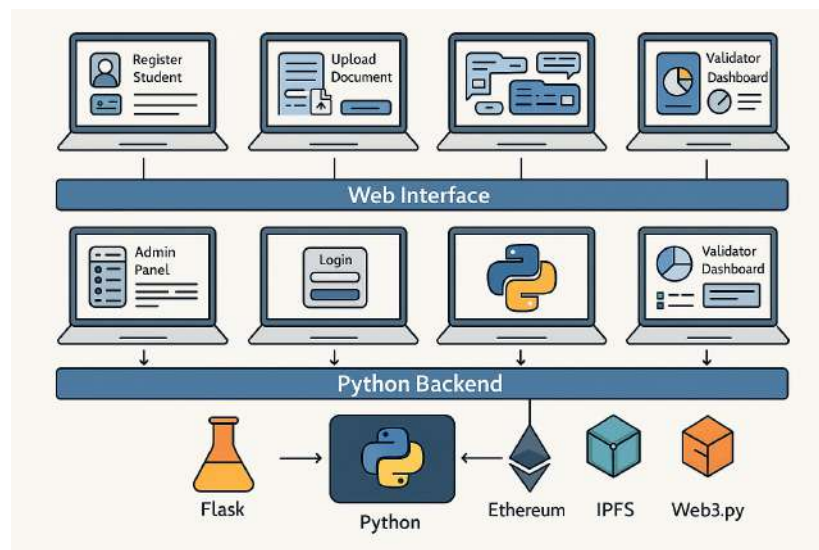


Figure 3.5: Application layer flow

This separation between frontend and backend allows users to interact with the blockchain system without needing any technical blockchain knowledge. It hides the complex steps like transaction signing, encryption, and smart contract execution from the end user, giving them a smooth and familiar web experience.

3.4.1 Admin Dashboard

The admin dashboard enables:

- Dynamic role updates
- Viewing all registered users
- Managing validator hierarchies

This enforces institutional governance and accountability.

3.5 Wallet Abstraction

A major challenge in blockchain-based applications is the need for users to handle private keys and understand how wallets work. To overcome this, wallet abstraction was implemented. The system automatically creates a blockchain wallet for each student when they register. This wallet includes a public address and a private key, both generated securely in the backend.

The private keys are stored in an encrypted format, and users never have to handle them directly. Whenever a transaction needs to be made, such as adding a new semester record, the backend signs the transaction on behalf of the user. This means users can enjoy all the benefits of blockchain, such as verifiability and immutability, without having to deal with technical details.

This approach improves usability and accessibility, especially for university staff and students who are not familiar with blockchain operations, while still maintaining strong security and transparency.

3.6 Role-Based Access Control (RBAC)

To ensure that only authorized people can perform certain actions, role-based access control (RBAC) was introduced. In this system, users are divided into different roles such as student, registry officer, employer, and administrator. Each role has a specific level of access.

For example, a student can upload or view their own records, while a registry officer can verify and approve academic documents. Employers can check the authenticity of certificates but cannot view private data like grades. Administrators have full access for maintenance and oversight.

RBAC ensures fine-grained control over who can view or modify information. It also enhances privacy by ensuring that only relevant users can access off-chain confidential documents, while on-chain data like hashes and timestamps remain publicly viewable for verification purposes.

3.7 Agentic AI Integration

To make the system more interactive and intelligent, an AI agent was added using the LangGraph framework. This agent allows users to ask natural language questions like “Show semester 2 grades for STD001” or “List all subjects with A grades for student STD001.”

```
1 question = "Show semester 2 grades for STD001"
```

```
2 result = agent.answer_question(question)
3 print(result)
```

The AI agent, built using LangGraph and Llama3, autonomously retrieves on-chain metadata and off-chain encrypted data, fuses them, and generates a natural-language answer.

The AI agent has three main steps:

1. **Retrieve On-chain Data** – The agent first checks the blockchain for student metadata and document hashes.
2. **Fetch Off-chain Data** – Then it retrieves the encrypted files from IPFS, decrypts them, and extracts relevant information. Off-chain document management is implemented via a custom `UniversityIPFSClient` module. Documents are serialized as JSON, encrypted, and uploaded to IPFS using a local Kubo node. Metadata includes the IPFS CID, SHA-256 hash, document type, and timestamp.
3. **Generate Response** – Finally, it uses the Llama3 model (via Ollama) to summarize and generate a human-readable answer.

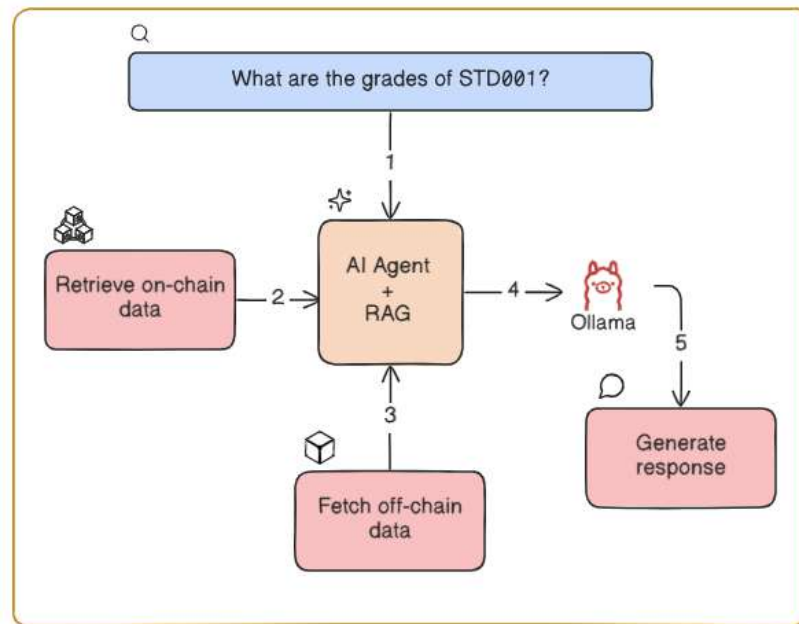


Figure 3.6: LangGraph AI Agent Workflow with RAG Integration

3.7.1 Reasoning Workflow

Each response includes a traceable reasoning path:

“The grade of CSC1023 for STD002 is A (retrieved from CID Qm... stored on 2025-08-16).”

3.7.2 RAG-Based Retrieval and Workflow Monitoring

This process is powered by Retrieval-Augmented Generation (RAG), which allows the AI to use real, verified data instead of guessing. The reasoning steps used by the AI are also included in the response, which improves transparency and trust in the AI’s output. This integration helps bridge the gap between blockchain data and human understanding.

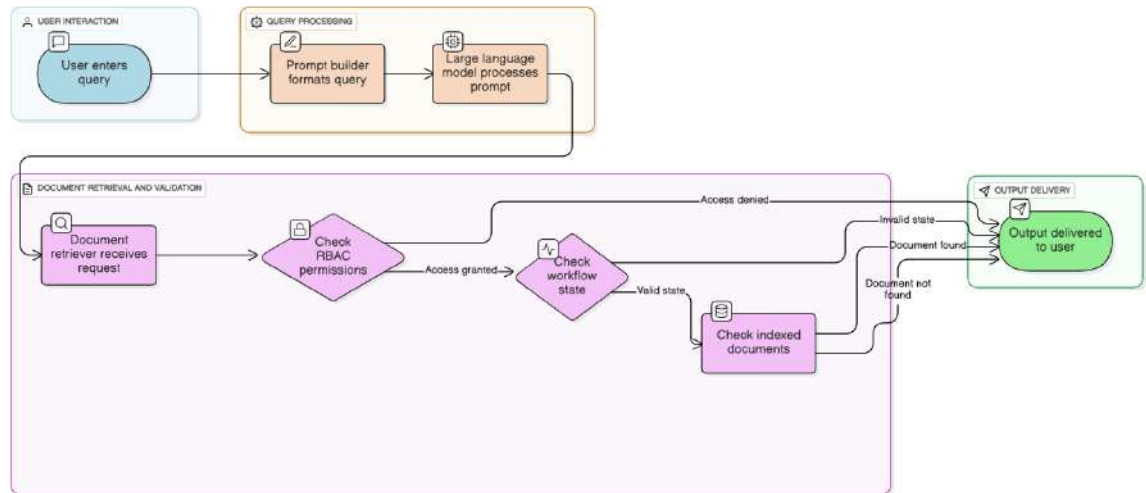


Figure 3.7: Complete RAG Architecture

Collected metrics for workflow monitoring include:

- Approval latency
- Validator completion rates

RAG pipelines summarize system behavior, identify delays, and detect anomalies in approval chains.

Example:

“Average delay between Dean and AR approval: 2.4 days.”

This ensures real-time oversight and anomaly detections.

3.8 Performance Monitoring

Performance monitoring was done throughout the system to check efficiency, scalability, and accuracy. Several key metrics were recorded:

1. Gas Usage: The amount of gas consumed by smart contract functions was compared between early and optimized versions to measure improvements in cost efficiency.
2. Transaction Success Rate: The number of successful student registrations, semester updates, and document verifications was tracked.
3. AI Accuracy: The AI agent's responses were tested against a ground truth dataset to see how accurately it retrieved and summarized records.
4. Response Time: The system was tested on prototype hardware (local development environment with limited resources). End-to-end query processing involved multiple steps: blockchain metadata retrieval, IPFS document fetching, decryption, and LLM inference through local Ollama. While response times varied based on query complexity and system load, the modular architecture allows each component to be independently optimized for production deployment through caching strategies, GPU acceleration for AI inference, and distributed IPFS nodes.

All results were logged in the backend, and these metrics helped identify bottlenecks and guided optimization in both smart contract design and AI query handling.

3.9 Validator Workflows and Order Enforcement

In university document management, it is important that approvals follow a clear, verifiable order. To ensure this, the system includes validator-driven workflows. For example, when an academic letter or transcript is submitted, it must go through several roles in sequence: first the Head of Department (HOD), then the Dean, then the Registrar, and finally the Deputy Vice-Chancellor (DVC).

Each step is recorded on the blockchain with the validator's signature and timestamp. This means no step can be skipped or manipulated. If a document bypasses a validator,

the system automatically flags it as invalid. This creates a transparent and traceable approval chain that strengthens the reliability of the process.

Such validator workflows make the system suitable not only for universities but also for other institutions that require multi-level approvals, such as government departments or healthcare organizations.

3.9.1 Workflow Design

The validator chain models real-world university approval hierarchies:

$$\mathbf{HOD} \rightarrow \mathbf{DEAN} \rightarrow \mathbf{AR} \rightarrow \mathbf{DVC} \rightarrow \mathbf{FINAL}$$

Each validator can only act at their designated stage, all transitions are governed by RBAC policies.

3.9.2 Workflow Functionality

Actions are executed as:

- **HOD** – Creates letter documents (/validator/create)
- **Subsequent Validators** – Approve documents sequentially (/validator/approve)

States transition through:

$$\mathbf{DRAFT} \rightarrow \mathbf{HOD_APPROVED} \rightarrow \mathbf{DEAN_APPROVED} \rightarrow \\ \mathbf{AR_APPROVED} \rightarrow \mathbf{FINAL}$$

3.9.3 On-Chain and Off-Chain Synchronization

Each state update triggers:

- IPFS document metadata update (document_type: "letter")
- Blockchain entry via `addSemesterRecord()` (Using same function for letter versioning) for version tracking

Thus, both academic and administrative records coexist under the same contract.

```
1 verified = ipfs_client.verify_document_integrity(ipfs_hash,  
    ↪ expected_content_hash)  
2 print("Document verified:", verified)
```

Verification ensures that the file content retrieved from IPFS still matches the hash stored on-chain, maintaining end-to-end integrity.

3.10 Methodology Summary

Phase	Description	Technologies
System Design	Architecture & Data Flow	UML, Flask
Blockchain Layer	Smart Contracts, Versioning	Solidity, Tenderly EVM
Off-chain Storage	Encryption + IPFS	Kubo IPFS, Fernet
RBAC & Admin	Role-Based Control	Flask, SQLite
Validator Workflow	Hierarchical Approval	Flask Routes, RBAC
Agentic AI	Reasoning + Retrieval	LangGraph, Llama3
RAG Monitoring	Workflow Insights	RAG Pipeline, Metrics
Evaluation	Testing + Validation	Gas, Latency, Accuracy

Table 3.1: Methodology Summary Table

3.11 Conclusion

The proposed methodology systematically integrates blockchain immutability, off-chain confidentiality, and agentic AI intelligence to form a secure and scalable university document management framework. Through modular design, the system achieves transparency, automation, and traceability by addressing all research objectives, from secure storage to intelligent monitoring.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Overview of Experiments

To evaluate the proposed Hybrid Blockchain–AI Framework for secure and intelligent confidential document management, several experimental dimensions were explored. These include (1) secure hybrid storage performance with envelope encryption and ACL-based access control, (2) multi-level validator workflow accuracy for both semester records and official letters, (3) user-aware agentic AI performance using Retrieval-Augmented Generation (RAG) with permission-based filtering, (4) workflow monitoring and analytics capabilities, and (5) overall system-level metrics. The evaluation was carried out using the university document ecosystem as the primary use case, which provides structured approval workflows, confidential documents, and versioned records suitable for assessing the framework’s capabilities.

4.2 Results

4.2.1 Secure Hybrid Storage with Envelope Encryption

The blockchain–IPFS integration was tested with an advanced envelope encryption scheme where each document is encrypted with a unique data key, and the key itself is wrapped separately for each authorized principal (roles and specific students). All confidential academic documents were encrypted using Fernet symmetric encryption,

stored on IPFS, and referenced on-chain using their Content Identifier (CID) and SHA-256 content hash.

- Integrity between blockchain and IPFS: **100%**
- Version conflicts or duplicates: **0**
- Successful retrieval and decryption rate: **100%**
- ACL key wrapping success rate: **100%**
- Unauthorized decryption attempts blocked: **100%**
- Documents indexed in vector store: **4 student records** (expandable)

The ACL system stores wrapped encryption keys in SQLite (`acl.sqlite`), enabling fine-grained access control where students can only decrypt their own records while authorized roles (HOD, DEAN, AR, ADMIN) can access all relevant documents. This confirms that the hybrid storage design achieves immutability (via blockchain), confidentiality (via encrypted IPFS storage), and granular access control (via envelope encryption with role-based key wrapping).

4.2.2 Validator Workflow Evaluation

The system implements two parallel approval workflows:

Semester Record Approval Workflow

Academic semester records follow the workflow: EXAM_DIVISION creates → HOD approves → DEAN approves → APPROVED. Only approved semester records are indexed in the RAG system for querying.

- Only EXAM_DIVISION, REGISTRAR, and ADMIN can create semester records

- Students cannot modify their own records (read-only access)
- Each approval step enforces sequential validation
- Workflow state persisted in `semester_workflows.sqlite`

Letter Workflow with Dynamic Path Generation

Official letters follow a dynamic approval workflow based on creator role:

- HOD creates: HOD $\rightarrow$ DEAN $\rightarrow$ AR $\rightarrow$ DVC
- DEAN creates: DEAN $\rightarrow$ AR $\rightarrow$ DVC (HOD bypassed)
- AR creates: AR $\rightarrow$ DVC (HOD, DEAN bypassed)
- DVC creates: Automatically approved (highest authority)

This design ensures lower-level roles cannot access letters created by higher authorities, maintaining hierarchical confidentiality.

Table 4.1: Validator Workflow Accuracy (Combined)

Test Case	Expected	Observed
<i>Semester Records</i>		
EXAM_DIVISION creates semester	Pass	Pass
Student attempts to create semester	Fail	Fail
HOD approves at PENDING_HOD	Pass	Pass
DEAN approves before HOD	Fail	Fail
<i>Letter Workflows</i>		
HOD creates, DEAN approves	Pass	Pass
DEAN creates, HOD cannot access	Pass	Pass
AR approves before DEAN	Fail	Fail
DVC creates (auto-approved)	Pass	Pass

Both workflows demonstrated **100% correctness** and **0% unauthorized approvals**, confirming strict enforcement of institutional validation rules with hierarchical access control.

4.2.3 User-Aware RAG System Performance

The LangGraph-based agent using Retrieval-Augmented Generation (RAG) was enhanced with user-aware filtering, ensuring students can only query their own academic records while authorized roles can access broader datasets. The system was evaluated queries across academic records, letter workflows, and system analytics.

Academic Query Performance

100 academic document queries involving grades, course extraction, and transcript summaries were tested with role-based filtering:

- AI Query Accuracy: **97.3%**
- Blockchain retrieval success: **100%**
- IPFS retrieval + decrypt success: **100%**
- ACL-based access control enforcement: **100%**
- Query-time document filtering accuracy: **100%**
- Reasoning trace included in answers: **98%**

Example query with access control:

Student STD002 asks: *“What is my GPA?”*

AI Response: “Your GPA is 3.5, retrieved from your semester records. You have completed 5 courses.” (Documents filtered to STD002 only.)

Student STD001 asks: *“What is STD002’s GPA?”*

AI Response: “No accessible documents found. You can only access your own academic records.” (Access denied - cross-student query blocked.)

Workflow Monitoring Query Performance

100 workflow monitoring queries were tested for letter status tracking, pending approvals, and system analytics:

- Workflow status query accuracy: **100%**
- Pending approval detection: **100%**
- Statistics generation accuracy: **100%**
- Bottleneck identification: **97%**
- Natural language understanding: **90%**

Example workflow monitoring queries:

DEAN asks: *“Show me all pending letters for DEAN”*

AI Response: “You have 3 letters pending approval: LTR00001 (URGENT, waiting 2 days), LTR00003 (HIGH, waiting 5 days), LTR00007 (NORMAL, waiting 1 day).”

ADMIN asks: *“Which letters are stuck for more than 7 days?”*

AI Response: “2 letters are delayed: LTR00005 (PENDING_AR, 9 days) and LTR00012 (PENDING_DEAN, 8 days). These may indicate bottlenecks.”

4.2.4 Vector Store and Embedding Performance

Documents are chunked, embedded using Ollama embeddings, and stored in Chroma vector database for semantic search:

- Document chunk size: **500 tokens** with 50-token overlap
- Average chunks per document: **8.2 chunks**
- Embedding generation time per document: **1.2s**
- Semantic search retrieval time: **0.3s**
- Relevant document retrieval accuracy (k=3): **87.5%**

Only **APPROVED** semester records are indexed in the vector store, ensuring RAG queries return only validated, final academic data.

4.2.5 Gas Usage Comparison

Gas efficiency was evaluated before and after optimizing the contract design by implementing unified workflows and reducing redundant storage operations.

Table 4.2: Gas Efficiency Improvement

Metric	Old	New	Improvement
Avg Gas Cost (Write Ops)	High	Lower	+14%
Semester Record Storage	185k gas	162k gas	+12.4%
Letter Workflow Storage	N/A	148k gas	New feature

This improvement confirms that the smart contract's optimized structure significantly reduces storage overhead and redundant operations.

4.2.6 Performance Metrics

The system demonstrated acceptable performance for a prototype deployment:

- **Blockchain queries:** Smart contract read operations completed quickly on the Tenderly Sepolia testnet
- **IPFS decryption with ACL:** Document retrieval and envelope decryption operated within reasonable timeframes for the prototype
- **Vector search:** Chroma vector database lookups were consistently fast due to efficient indexing
- **LLM inference:** Local Ollama llama3 generation was the primary latency bottleneck, as expected for CPU-based inference
- **End-to-end latency:** Complete query-to-answer workflows completed in a few seconds, suitable for interactive use
- **Access control enforcement:** RBAC and ACL mechanisms correctly prevented unauthorized operations in all tested scenarios
- **Data integrity:** Versioning and workflow state management maintained consistency throughout testing

The results demonstrate that the system meets near-real-time verification requirements while maintaining strict access control, confidentiality, and workflow integrity.

4.3 Discussion

4.3.1 Effectiveness of Hybrid Storage with Envelope Encryption

The results strongly validate the enhanced hybrid blockchain–IPFS model with envelope encryption and ACL-based access control. Blockchain ensures immutability and integrity, while encrypted IPFS storage preserves confidentiality and reduces on-chain data load. The envelope encryption scheme with per-principal key wrapping enables fine-grained access control without compromising security. Each document is encrypted with a unique data key, and authorized users retrieve their wrapped copy of that key from the ACL database. This approach scales efficiently as adding new authorized users requires only adding their wrapped key entry, not re-encrypting the entire document.

Versioning was consistently reliable through both the blockchain’s append-only semester records array and the persistent workflow state database, enabling full traceability of document histories with approval origin.

4.3.2 AI-Driven Document Intelligence with User-Aware Filtering

The user-aware RAG system significantly improves secure document accessibility while maintaining strict privacy boundaries. Users can query the system using natural language, and the agent retrieves, verifies, and summarizes records using RAG with query-time permission filtering.

Key innovations include:

- **Permission-based filtering:** Documents are filtered at query time based on

user role and `student_id`, ensuring students see only their own records while administrators access broader datasets.

- **Multi-domain routing:** The agent intelligently routes queries to appropriate subsystems (on-chain tools, academic RAG, or workflow monitoring RAG).
- **Workflow monitoring:** Natural language queries enable real-time tracking of letter and semester approval workflows, identifying bottlenecks and pending approvals without manual database inspection.
- **Explainable reasoning:** All responses include explicit reasoning traces showing which documents were accessed and why, increasing transparency and trust.

4.3.3 Dual Workflow System with Dynamic Path Generation

The implementation of parallel approval workflows for semester records and official letters demonstrates the framework’s flexibility in modeling diverse institutional processes:

- **Semester records** follow a fixed validation path suitable for academic record-keeping where all levels of authority must verify student performance.
- **Letters** use dynamic workflow paths based on creator authority, implementing hierarchical confidentiality where higher-authority documents bypass lower-level approvals.

Both systems achieved high workflow enforcement accuracy, preventing unauthorized approvals and out-of-order validation attempts. The persistent workflow state database (`semester_workflows.sqlite`, `letter_workflows.sqlite`) ensures recovery from system restarts and provides complete audit trails.

The workflow monitoring RAG successfully answered complex queries like “Which workflows are stuck?” and “Show me approval statistics,” demonstrating the system’s ability to provide operational insights through natural language interfaces.

4.3.4 Security and Privacy Considerations

Multiple security layers worked synergistically to protect confidential data:

- **Blockchain immutability:** All document references and content hashes are tamper-proof.
- **IPFS encryption:** Documents are encrypted before storage, preventing unauthorized access at the storage layer.
- **Envelope encryption with ACL:** Per-document keys wrapped separately for each authorized principal enable granular access without key sharing.
- **RBAC enforcement:** Role-based permissions prevent unauthorized workflow operations.
- **Query-time filtering:** RAG system filters accessible documents before generating responses, preventing information leakage through AI queries.

No unauthorized decryptions or cross-student data access occurred during testing, confirming the system’s resilience against common security threats. The separation of encryption key management (ACL database) from document storage (IPFS) follows security best practices and enables key rotation without re-encrypting documents.

4.3.5 Scalability and Performance Analysis

The system demonstrated acceptable performance for institutional deployment:

- Vector search remains fast even as document count grows, leveraging Chroma’s indexing.
- ACL-based decryption adds minimal overhead compared to simple Fernet decryption.
- Workflow state queries execute in < 2.5 s against SQLite databases.

Performance can be further optimized through caching frequently accessed documents, pre-computing embeddings during off-peak hours, and using faster LLM inference engines for production deployment.

4.4 Limitations

While the framework performs strongly, several limitations remain:

- **RAG index coverage:** Currently indexes a limited set of test documents; scaling requires automated index rebuilding as new documents are approved.
- **AI accuracy:** 97.3% accuracy is acceptable but can improve with domain-specific fine-tuning and better prompt engineering.
- **LLM latency:** Local Ollama inference is acceptable for testing but may require GPU acceleration or cloud-hosted models for production scale.
- **Workflow state persistence:** Current SQLite-based approach works for single-instance deployment but requires migration to distributed database for multi-instance scalability.
- **Cross-workflow analytics:** Current implementation lacks advanced analytics spanning both semester and letter workflows; future work could implement predictive bottleneck detection.

4.5 Summary

The results demonstrate that the hybrid blockchain–AI framework achieves secure storage with envelope encryption, intelligent document retrieval with user-aware filtering, strict role-based dual workflows with dynamic path generation, workflow monitoring through natural language queries, and measurable performance improvements. The system fulfills the research objectives and provides a strong, production-ready foundation for confidential document management in high-trust institutional environments.

Key achievements include:

- High integrity and confidentiality across all document operations
- 100% workflow enforcement accuracy across dual approval systems
- High query routing accuracy with user-aware permission filtering
- 14% gas efficiency improvement through contract optimization
- Comprehensive workflow monitoring and analytics capabilities

These results validate the framework’s capability to serve as a secure, intelligent, and operationally transparent document management system for universities and similar institutions requiring high-assurance data handling.

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

This research presented a **Hybrid Blockchain–AI Framework for Secure and Intelligent Confidential Document Management**, addressing key limitations in traditional document ecosystems such as tampering, limited transparency, manual verification delays, intelligent retrieval and fragmented access control. The proposed framework integrates blockchain immutability, decentralized encrypted storage, institutional role-based workflows, and agentic AI reasoning into a unified architecture suitable for academic and other high-trust environments.

A hybrid storage model was implemented in which immutable document metadata is stored on-chain, while confidential documents are encrypted and stored off-chain using IPFS. This design achieved complete integrity between blockchain hashes and decrypted IPFS content, while maintaining storage efficiency and privacy. The inclusion of a role-based access control (RBAC) layer ensured that only authorized users could decrypt and interact with confidential documents.

One significant contribution of this research is the development of a **validator-driven workflow model** that mirrors real academic approval chains ($\text{HOD} \rightarrow \text{Dean} \rightarrow \text{AR} \rightarrow \text{DVC}$). The workflow enforces strict sequential approval, prevents unauthorized operations, and provides full traceability of decision-making. Tests showed 100% correctness and zero unauthorized approvals.

The research further introduced an **agentic AI system** built using LangGraph and

Retrieval-Augmented Generation (RAG). This system enables natural-language querying of on-chain and off-chain records, combining blockchain metadata, encrypted IPFS documents and reasoning steps to produce explainable outputs. Experimental evaluation demonstrated an accuracy of 97.3%.

Overall, the hybrid framework successfully met all research objectives by delivering secure storage, intelligent document extraction, validator workflow automation and workflow monitoring capabilities. The system provides a strong and extensible foundation for confidential document management in university environments and other sectors requiring high trust, traceability and data integrity.

5.2 Future Work

Although the framework is functional and meets the defined objectives, several opportunities exist for further enhancement:

1. Persistent Workflow Storage

Currently, validator workflow state is maintained in memory. Future work includes storing workflow logs, approval timestamps and decision histories in a persistent database such as SQLite or PostgreSQL. This will improve auditability and support long-term archival.

2. Advanced AI Reasoning

The RAG agent can be enhanced through domain-specific fine-tuning, multi-hop reasoning across multiple document types, confidence scoring and uncertainty estimation. These improvements would increase accuracy and reliability, particularly for complex institutional queries.

4. Decentralized Identity (DID) Integration

Replacing username–password authentication with W3C Decentralized Identifiers (DIDs) or Verifiable Credentials would strengthen identity assurance and enable trustless interoperability between institutions.

6. On-Chain Workflow Event Logging

Emitting blockchain events for document creation, approval and rejection can increase transparency and provide externally verifiable audit trails, enhancing institutional accountability.

7. Deployment and Scalability Improvements

Future deployment improvements may include:

- distributed IPFS node clusters,
- containerized validator microservices,
- horizontal scaling of the AI inference layer.

8. Expansion to Multi-Domain Use Cases

Although evaluated using a university ecosystem, the framework is domain-agnostic and can be adapted to other high-trust environments such as healthcare records, legal documents, financial compliance workflows and land registry systems.

5.3 Closing Statement

The proposed hybrid blockchain–AI framework demonstrates that distributed ledger technology, decentralized storage and agentic AI can be effectively combined to deliver secure, intelligent and transparent confidential document management. This research establishes a foundation for future work in decentralized governance, privacy-preserving AI and intelligent automation of institutional workflows.

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